

Immersive Media and Virtual Accessibility in Healthcare

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1 INTRODUCTION

Healthcare is a field where the unique affordances of virtual reality can be profoundly effective. Presence and immersion and empathy are powerful tools in a context that is both emotionally complex and where the stakes are particularly high. Uses of virtual reality in the medical field are as wide ranging as providing pain relief without drugs to creating unique group therapy opportunities. The goal of this panel is to discuss how expanded accessibility can offer unique benefits to populations that have the most to gain. Young people suffering from isolation, wheelchair users adapting to new limits on their mobility, and those who simply want access to the same experiences that are available to others.

Often the people who have the most to gain from novel strategies are the last to benefit from them. Current technology is well designed to serve the general population. For people with atypical needs, emerging technologies can offer expanded opportunities and access, but only if we build with those concerns in mind. Where industry has a narrow focus on broad adoption, innovators and clinicians can focus on deep adoption, where the population that benefits is not as large but where those benefits can be transformative.

Asher Marks and Kimberly Hieftje run the PedsXR lab at Yale which use virtual reality to help empower youth and to provide them therapeutic tools for overcoming illness and isolation. Yetisa Tuakli, founder of the Sports Equity Lab, is working to find new ways that virtual reality can communicate the challenges faced by disabled athletes, while hoping to provide them new tools for therapy and training. Grzegorz Bednarski is creating tools such as WalkinVR that allow people with motor impairment to experience commercial VR products such as Beat Saber and other games. The value of a panel like this is that unique populations have unique needs and expanding accessibility is an incredibly difficult task. It requires a deep understanding of those needs and the communities

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they are endemic to, while also demanding that providers understand the more general trends within the industry. It does little good to make accessible a technology that is outdated by the time it is ready for clinical use. Combining clinicians like Dr. Marks and Dr. Wosornu with developers and designers like Dr. Hieftje and Grzegorz Bednarski is a way to ensure that the needs of specific communities are a part of clinical projects from the very foundation of the development process. What are the best strategies for ensuring compatibility with tools such as WalkinVR and what are the platforms that provide the right combination of support and privacy?

2 STRUCTURE

Each participant will offer a short 5 minute introduction of their background and highlight a single project that best describes their relevant experience. The moderator will present a series of questions to the group and allow members to provide individual answers to these questions. Once these general questions have been answered there will be another section that allows the panelists to frame questions to each other.

3 MODERATOR

Justin Berry is a Critic at the Yale School of Art and Core Faculty at the Yale Center for Collaborative Arts and Media. His work was recently included in "How to Win at Photography" at the Fotomuseum Winterthur. Issues of Artforum, Art Review, and Frieze have featured his work. As the Principal Investigator at Yale's Blended Reality Lab he is currently working on a concrete interface for virtual reality that is designed to work with wheelchairs.

4 PANELISTS

Asher Marks, MD is an Assistant Professor of Pediatrics in the section of Pediatric Hematology/Oncology. He is the director of the Pediatric Neuro-Oncology program and Adolescent/Young Adult Oncology program. His research involves clinical trials of early phase therapeutics as well as the integration of novel digital interventions into the supportive care of these patient populations.

Kimberly Hieftje, PhD, is an Assistant Professor of Pediatrics, director of the play4REAL XR Lab and co-director of XRPeds at Yale. For the past 13 years, her research has focused on the development, evaluation, and implementation of health and education interventions for youth that utilize XR and game technology.

Yetisa A. Tuakli-Wosornu, MD, MPH is an Associate Professor at Yale School of Public Health and University of Pittsburgh. Board-certified in Physical Medicine and Rehabilitation, specializing in spine and sports medicine. Dr. Tuakli directs the Sports Equity Lab—a research group focused on dismantling inequities in sport. A long jumper who represented the Ghana national team until 2016, Dr. Tuakli was the International Paralympic Committee's inaugural

welfare officer, is a member of the International Blind Sports Federation medical committee, and is Chair of the International Society for Physical Medicine and Rehabilitation task force on physical activity for persons with disabilities.

Grzegorz Bednarski is a developer and creator of WalkinVR, a plug-in to SteamVR, that provides people with impaired mobility the option to customize the interface for virtual reality to suit their individual needs.